

Date	11 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap 5, JavaScript	Provides a responsive and user-friendly interface for applicants and bank officers to enter loan details and view prediction results across desktop and mobile devices.
2	Backend / Server-Side	Python, Flask	Flask provides a lightweight and efficient framework for handling application logic and integrating the trained machine learning model with the web interface.
3	Database / Data Storage	SQLite	Stores applicant information, loan application records, and prediction outcomes. SQLite is easy to configure and suitable for academic-scale projects.
4	Cloud / Hosting / Deployment	IBM Cloud	Offers a secure and scalable environment to deploy the application, allowing users to access the system through the internet with minimal infrastructure management
5	Version Control & CI/CD	Git, GitHub	Facilitates source code management, version tracking, collaboration among team members, and maintains project history throughout development.

6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Joblib, Jupyter Notebook	These tools are used for data preprocessing, model training, performance evaluation, model serialization, and experimentation during the machine learning development process.
---	---------------------------------------	---	--

